

Julian Triveri

Mixed Reality Software Engineer & Technical Artist

trev3d.com · github.com/trev3d · linkedin.com/in/jtriveri

Software engineer and technical artist **specializing in mixed reality systems** and **3D graphics** on **resource-constrained hardware**. Creative problem solver with a knack for **engineering around hardware and software limitations**. Deep platform-level expertise on Meta Quest, including undocumented APIs, **real-time environment reconstruction pipelines**, and **mass-collocated mixed-reality (30+ users in the same space)**. Owns the full stack of real-time 3D, from **low-level shaders to high-fidelity art** optimized for mobile GPUs. Work featured on *UploadVR*, *Heise Online*, and *80 Level*. Won **Judges Choice, 2025 Meta Horizon Start Developer Competition**. Recent graduate from Purdue University.

EXPERIENCE

Anaglyph — Founder, Software Engineer & Graphics Programmer

West Lafayette, IN

Arena-scale multiplayer mixed reality games on Meta Quest. Open-sources technologies for other developers.

Mar 2023 – Present

- **Built real-time environment reconstruction on Meta Quest** using Depth API and GPU compute shaders—constructing a 3D scan from depth frames, SDF integration, surface-net meshing, and automatic LOD generation for collision and visual occlusion **with zero prior room setup**. Solved a core platform limitation and significant UX pain-point unaddressed on Quest. Published open source.
- **Developed and published camera access work-around on Quest** via Android media projection API, enabling custom computer vision (e.g. fiducial marker tracking). Published open source.
- **Engineered multi-marker collocation system** scaling multiplayer play areas from a single room to entire buildings, overcoming standard Quest tracking and shared anchor limitations.
- Created *Lasertag* arena-scale first-person shooter leveraging previously mentioned technologies. **Won Judges Choice, 2025 Meta Horizon Start Competition**.
- Created voice-activated wizard dueling game for in-person Purdue University student orientation; **played by hundreds of users per day**, lines with 30+ minute waits for 2-minute matches.
- Prototyped **multithreaded iterative-closest-point solver** comparing live environment meshes against depth point clouds to correct tracking drift in large environments on Meta Quest.

Purdue University Envision Center — Research SWE & Technical Artist

West Lafayette, IN

Interactive educational MR apps for university STEM students. Full ownership: 3D art, shaders, simulation, UX.

Oct 2021 – Present

- Built **collocated MR classroom for 30 simultaneous users** in Quest headsets. Used undocumented Meta Quest APIs to **reduce class setup time from 30 min to under 2 min** using camera access & fiducial marker tracking work-around.
- Created **photorealistic virtual nuclear engineering lab simulating equipment** on low-spec devices (Chromebooks/tablets via WebGL)—25K total tri scene, 22 draw calls, fully simulated signal processing at 500hz on a single CPU core. Expanded registration for courses previously constrained by physical lab space/equipment needs.
- Implemented scientific visualizations (volumes, point clouds) using **HLSL fragment and compute shaders on standalone MR headsets** across nuclear engineering, radiopharmaceutical safety, industrial hygiene, meteorology, and astrophysics.

EDUCATION

Purdue University — B.A. Animation & Visual Effects

Aug 2021 – May 2026

PRESS & RECOGNITION

- UploadVR – “[Another Developer Found an Easier Way to Access Quest 3’s Passthrough Cameras](#)” (Aug 2024)
- Purdue University – “[Computing, Graphics Senior Creates Innovations That Work in Practice, Not Just Theory](#)” (Jan 2025)
- UploadVR – “[A Developer Implemented Continuous Scene Meshing on Quest 3](#)” (May 2025)
- Heise Online – “[‘Lasertag’ anywhere and anytime: mixed reality opens up new possibilities](#)” (Aug 2025)
- UploadVR – “[Wizardtag Is a Spellcasting Version of The Amazing Quest 3 Game Lasertag](#)” (Oct 2025)
- Meta Horizon Start Developer Competition – [Judges Choice Award](#) (Dec 2025)

TECHNOLOGIES

Platforms: Meta Quest, Android/WebGL via Unity.

Languages: C#, HLSL

Frameworks: Unity, Job System & Burst Compiler, Meta Quest SDK, Unity Netcode for GameObjects

Graphics & Simulation: GPU compute shaders, volumetric rendering, point clouds, real-time reconstruction via depth

3D Art: Hard-surface modeling, PBR shading, draw call budgeting, lightmap baking (Blender, Substance 3D Painter, Marmoset Toolbag)

Tools: Git, Claude Code, Motive/OptiTrack